

# Wenqin Chen

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## RESEARCH INTERESTS

Machine learning for quantum many-body physics (neural quantum states); quantum geometry of superconductivity and of phonon magnetism; topological phases, skyrmion crystals, and quantum anomalous Hall states; topological quantum computing with Majorana zero modes.

## EDUCATION

**University of Washington**, Seattle, WA 2021 – 2027 (expected)

*PhD in Physics. Advisors: Di Xiao, Ting Cao*

- Thesis: Machine Learning and Quantum Many-Body Theory for Condensed Matter Physics

**Xi'an Jiaotong University**, Xi'an, China 2016 – 2021

*BS in Physics (Honors)*

- Outstanding Undergraduate Thesis Award

**University of California, Berkeley** 2019

*Exchange student in Physics*

## RESEARCH EXPERIENCE

**University of Washington**, Seattle, WA 2021 – Present

*Graduate Research Assistant. Advisors: Di Xiao, Ting Cao*

- Neural quantum states: built a transformer wavefunction ansatz that solves the continuum many-body Schrödinger equation in JAX, optimized by variational Monte Carlo with natural-gradient (stochastic reconfiguration / MinSR) updates, warm-started from Hartree-Fock and benchmarked against exact diagonalization (matched the exact ground state within 0.7 mHa, recovering 96% of the correlation energy, at two electrons); production runs at up to 18 electrons on H200 GPUs (UW Tillicum) and Google Cloud Platform.
- Open-sourced the solver as a reusable, CI-tested (GitHub Actions) NQS module (attention ansatz, Metropolis MCMC sampler, Ewald Coulomb, MinSR optimizer) with an exact-oracle pytest suite: [github.com/wenqin-chen/continuum-nqs](https://github.com/wenqin-chen/continuum-nqs).
- Quantum geometry of superconductivity: developed the Cooper-pair quadrupole framework showing that Berry curvature and the quantum metric set a lower bound on the pair size in flat-band superconductors [arXiv:2605.03133]; applied to rhombohedral graphene, where the predicted pair sizes match experimentally inferred coherence lengths.
- Phonon magnetism: built gauge-theory and geometric descriptions of giant phonon magnetic moments in Dirac materials [PRB 111, 035126; arXiv:2505.09732; arXiv:2605.06983], with predictions guiding magneto-Raman experiments by collaborators at Los Alamos (CINT) and UC Irvine.
- Co-developed the theory of adiabatic pumping of orbital magnetization by spin precession [PRL 134, 176702].

**Los Alamos National Laboratory (T-4 / CNLS)**, Los Alamos, NM 2023 – 2024

*Research Intern (two summer internships). Mentor: Shi-Zeng Lin*

- Delivered two first-author papers on phonon magnetism in consecutive summers [PRB 111, 035126; arXiv:2505.09732].
- In an ongoing UW-LANL collaboration, built a self-consistent Hartree-Fock framework to search for skyrmion crystals and quantum anomalous Hall order in strongly correlated two-dimensional electron systems; open-sourced the ~7k-line, 77-test solver core ([github.com/wenqin-chen/hubbard-meanfield](https://github.com/wenqin-chen/hubbard-meanfield)) and ran large-scale parameter-scan campaigns on the DOE NERSC Perlmutter supercomputer (SLURM, 128 cores/task).

**Lawrence Berkeley National Laboratory**, Berkeley, CA 2019

*Research Intern. Mentor: Joseph Orenstein*

- Developed numerical simulations in direct collaboration with the experimental team to interpret optical birefringence measurements of the three-state nematic antiferromagnet Fe<sub>1</sub>/3NbS<sub>2</sub>; co-authored the resulting paper [Nature Materials (2020)].

**Xi'an Jiaotong University**, Xi'an, China 2019 – 2021

*Undergraduate Researcher. Mentor: Jie Liu*

- Simulated non-Abelian braiding of Majorana zero modes in a tetron device and showed that geometric-phase cancellation restores braiding-time-independent statistics in the presence of an Andreev bound state, the most common experimental contaminant in Majorana devices; first-author paper [PRB 105, 054507].
- Co-developed a minimal quantum-dot braiding protocol with electrical fermion-parity readout [SCPMA 64, 117811].

## PUBLICATIONS

**Nonadiabatic Theory of Phonon Magnetic Moments in Insulators and Metals** 2026

Haoran Chen, **Wenqin Chen**, Kaijie Yang, Ting Cao, Di Xiao · *arXiv:2605.06983 (submitted to Physical Review Letters)*

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| <b>Quantum Geometric Quadrupole of Cooper Pairs</b>                                                                                                                                                                                   | 2026 |
| <i>Wenqin Chen</i> , Kaijie Yang, Ting Cao, Shi-Zeng Lin, Jiabin Yu, Di Xiao · <i>arXiv:2605.03133 (submitted to Physical Review Letters)</i>                                                                                         |      |
| <b>Geometric Origin of Phonon Magnetic Moment in Dirac Materials</b>                                                                                                                                                                  | 2025 |
| <i>Wenqin Chen</i> , Xiao-Wei Zhang, Ting Cao, Shi-Zeng Lin, Di Xiao · <i>arXiv:2505.09732</i>                                                                                                                                        |      |
| <b>Adiabatic Pumping of Orbital Magnetization by Spin Precession</b>                                                                                                                                                                  | 2025 |
| Yafei Ren, <i>Wenqin Chen</i> , Chong Wang, Ting Cao, Di Xiao · <i>Physical Review Letters</i> 134, 176702                                                                                                                            |      |
| <b>Gauge Theory of Giant Phonon Magnetic Moment in Doped Dirac Semimetals</b>                                                                                                                                                         | 2025 |
| <i>Wenqin Chen</i> , Xiao-Wei Zhang, Ying Su, Ting Cao, Di Xiao, Shi-Zeng Lin · <i>Physical Review B</i> 111, 035126                                                                                                                  |      |
| <b>Non-Abelian Statistics of Majorana Zero Modes in the Presence of an Andreev Bound State</b>                                                                                                                                        | 2022 |
| <i>Wenqin Chen</i> , Jiachen Wang, Yijia Wu, Jie Liu, X. C. Xie · <i>Physical Review B</i> 105, 054507                                                                                                                                |      |
| <b>Minimal Setup for Non-Abelian Braiding of Majorana Zero Modes</b>                                                                                                                                                                  | 2021 |
| Jie Liu, <i>Wenqin Chen</i> , Ming Gong, Yijia Wu, X. C. Xie · <i>Science China Physics, Mechanics &amp; Astronomy</i> 64, 117811                                                                                                     |      |
| <b>Observation of Three-State Nematicity in the Triangular Lattice Antiferromagnet Fe<sub>1/3</sub>NbS<sub>2</sub></b>                                                                                                                | 2020 |
| Arielle Little, Changmin Lee, Caolan John, Spencer Doyle, Eran Maniv, Nityan L. Nair, <i>Wenqin Chen</i> , Dylan Rees, Jörn W. F. Venderbos, Rafael Fernandes, James G. Analytis, Joseph Orenstein · <i>Nature Materials</i> 19, 1062 |      |

## TALKS AND PRESENTATIONS

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- Quantum Geometric Quadrupole of Cooper Pairs in Flat-Band Superconductors** — APS Global Physics Summit, Denver (*contributed*), Mar 2026
- Quantum Geometric Quadrupole of Cooper Pairs in Flat-Band Superconductors** — MEM-C IRG-2 Seminar, University of Washington (*invited seminar*), Nov 2025
- Chiral Phonons** — LANL–University of Michigan Joint Workshop on Quantum Materials, Los Alamos, NM (*poster*), Jul 2025
- Gauge and Geometric Origins of Phonon Magnetism in Dirac Materials** — Oka Group Seminar, Institute for Solid State Physics, University of Tokyo, online (*invited seminar*), May 2025
- Geometric Theory of Phonon Magnetic Moment in Dirac Materials** — APS Global Physics Summit, Anaheim (*contributed*), Mar 2025
- Phonon Magnetic Moments in Cd<sub>3</sub>As<sub>2</sub>: Theory Guiding the Magneto-Raman Experiment** — presentations to the experimental collaboration, Los Alamos National Laboratory, online, Sep 2024
- Theory of Giant Phonon Magnetic Moment in Doped Dirac Semimetals** — APS March Meeting, Minneapolis (*contributed*), Mar 2024
- Dynamically Driven Orbital Magnetization Beyond the Adiabatic Limit** — APS March Meeting, Las Vegas (*contributed*), Mar 2023

## TEACHING

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- Teaching Assistant, Department of Physics, University of Washington (Autumn 2021, Spring 2022)

## AWARDS AND FELLOWSHIPS

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- Clean Energy Institute Graduate Fellowship, University of Washington (2024–2025)
- Thouless Institute for Quantum Matter Graduate Fellowship, University of Washington (2021)
- Google Cloud Research Credits Award (2026)
- Outstanding Undergraduate Thesis Award, Xi'an Jiaotong University (2021)

## TECHNICAL SKILLS

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**Machine learning:** Deep learning (neural networks: transformer/attention architectures, neural quantum states), variational Monte Carlo, natural-gradient optimization (stochastic reconfiguration / MinSR), score-function (REINFORCE-style) gradient estimators, ablation studies, training stabilization (outlier-robust gradient clipping, learning-rate schedules, warm starts, early stopping)

**Programming:** Python (NumPy, SciPy, pandas, Matplotlib), JAX (Flax, Optax), PyTorch, Bash/Linux, Git, pytest

**Scientific computing:** Solid-state electronic-structure methods (exact diagonalization, self-consistent-field Hartree-Fock), Bogoliubov-de Gennes / Nambu modeling of superconductor-semiconductor (Majorana) nanowire devices, Berry-curvature and transport (Kubo) calculations, Markov chain Monte Carlo (Metropolis sampling), Ewald summation, numerical linear algebra

**Infrastructure:** GPU training in JAX (H200), SLURM/HPC job orchestration (NERSC Perlmutter, UW Tillicum), Google Cloud Platform, Azure, CI (GitHub Actions), reproducible research pipelines

## SERVICE AND ACTIVITIES

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- Organizer, condensed-matter-theory group meetings, University of Washington (2025 – Present)
- Co-organizer, Summer Research Symposium for condensed matter physics, University of Washington (2025)
- K-12 STEM outreach: Clean Energy Institute Solar Derby, Nelsen Middle School (2025); UW Engineering Discovery Days (2024)
- Member, American Physical Society (Division of Condensed Matter Physics)